

# Rat *Pnlip* (P27657) — Secondary Lipolysis & Fatty-Acid Pathway Scope

**Hypothesis** slug: secondary-lipolysis-and-fatty-acid-pathway-scope **Focus** type: function\_assignment **Target:** Rattus norvegicus *Pnlip* / Pancreatic triacylglycerol lipase / UniProt P27657 / EC 3.1.1.3

## Executive Judgment

**Verdict:** REFUTED / OVER-ANNOTATED for the secondary functions in the seed hypothesis.

The seed hypothesis asks whether rat pancreatic lipase (*Pnlip*) genuinely (a) has phospholipase A1 activity, (b) has lipoprotein lipase activity, (c) participates in HDL remodeling or fatty-acid biosynthesis, or (d) positively regulates triacylglycerol lipase activity. Evaluating each separately against the actual UniProt/GO record and primary literature, none of these represent a directly-supported, core function of the classical pancreatic lipase gene product:

- The **only experimentally supported (IDA) molecular functions** for P27657 are **triacylglycerol lipase activity (GO:0004806)** and its parent **lipase activity (GO:0016298)**, localized to the **extracellular region (GO:0005576, IDA)**. Its catalytic reaction is  $TG + H_2O \rightarrow DAG + \text{fatty acid (sn-1,3 specific)}$ , i.e. a hydrolase/catabolic enzyme.
- **All four** flagged terms that the seed treats as suspect — **glycerophospholipid phospholipase A1 activity (GO:0008970)**, **lipoprotein lipase activity (GO:0004465)**, **HDL particle remodeling (GO:0034375)**, and **fatty acid biosynthetic process (GO:0006633)** — carry evidence code **IBA:GO\_Central**, i.e. **phylogenetic propagation (PAINT) from the PANTHER PTHR11610 lipase-family ancestral node** (the node referenced in the seed as PTN000906454). They are family-level inferences, **not direct evidence for classical pancreatic lipase**.
- The fifth flagged term, **positive regulation of triglyceride lipase activity (GO:0061365)**, traces (rat ISO ← mouse ISO ← human PNLIP IDA

to a paper that characterizes pancreatic lipase's **own** colipase-dependent interfacial activation — i.e., it is a **mis-mapping** of the enzyme's own catalysis onto a "regulation-of" term, not evidence that Pnlip regulates a separate triglyceride lipase.

**Most important caveats.** (1) "Refuted as a core function" is not the same as "zero activity in vitro" — classical pancreatic lipase is an sn-1,3 carboxyl esterase, so any residual phospholipid hydrolysis would be positional PLA1 chemistry at sn-1, but the *rate is negligible* and non-physiological. This is confirmed **rat-specifically** by the seed's own citation

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(Jennens & Lowe, *J Lipid Res* 1995), which states rat PLRP2 "could hydrolyze phospholipids, a **substrate poorly hydrolyzed by PL**" (PL = classical colipase-dependent pancreatic lipase). (2) The retinyl-ester activities (GO:0047376 ISS, GO:0050253 ISO) are separate secondary functions not part of the seed and are left as non-core.

**Iteration-2 verification (QuickGO with/from provenance).** All four flagged terms carry evidence IBA under GO\_REF:0000033 (PAINT) and share ancestral node **PANTHER:PTN000906454**; I resolved their "with/from" seed proteins to precise identities, proving they are inherited from **vascular-lipase paralogs**, not from pancreatic-lipase evidence: - GO:0004465 lipoprotein lipase activity ← **P06858 (human LPL) + P11151 (bovine LPL)** - GO:0008970 phospholipase A1 activity ← **P06858 (LPL), P11150 (hepatic lipase LIPC), Q9Y5X9 (endothelial lipase LIPG)** - GO:0034375 HDL remodeling ← **P11150 (LIPC), Q9Y5X9 (LIPG)** - GO:0006633 fatty acid biosynthetic process ← **P06858 (LPL), P11150 (LIPC), P11151 (bovine LPL)**

Conversely, the **core** term GO:0004806 triacylglycerol lipase activity is backed by **IDA**

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+ EXP

P 10769148

and GO:0005576 extracellular region by IDA

P 17010228

P 8967484

**Iteration-3 verification (GO:0061365 traced to origin).** The fifth flagged term, positive regulation of triglyceride lipase activity, resolves along an orthology chain: rat P27657 (ISO:RGD) ← mouse *Pnlip* Q6P8U6 (ISO, GO\_REF:0000119) ← **human PNLIP P16233 (IDA,**

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P 9631512

That root paper (Yang & Lowe, *Protein Expr Purif* 1998) is a **self-characterization** of recombinant human pancreatic lipase — it reports the enzyme "was inhibited by bile salts, required colipase for activity, and demonstrated **interfacial activation**." It does **not** show Pnlip regulating a *separate* triglyceride lipase. GO:0061365 therefore appears to be a **mis-mapping of the enzyme's own interfacial activation/colipase dependence** onto a "regulation-of" term; the immediate event is the enzyme's own catalysis (GO:0004806). Lead: remove or re-map GO:0061365 to the core catalytic term, pending curator review.

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## Evidence Matrix

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| Citation                                                                                                          | Evidence type                                          | Supports/<br>Refutes/<br>Qualifies  | Claim tested                                                                 | Key finding                                                                                                                                                                                                                                                       | Context                                                                               | Con<br>&<br>limi                              |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------|
| UniProt<br>P27657<br>(database)                                                                                   | Database<br>record                                     | Qualifies<br>(frames all)           | Actual GO/EC<br>of rat Pnlip                                                 | Core IDA MFs =<br>triacylglycerol<br>lipase<br>(GO:0004806) +<br>lipase<br>(GO:0016298); CC<br>= extracellular<br>(GO:0005576,<br>IDA). All 4 flagged<br>secondary terms<br>are <b>IBA</b> ;<br>GO:0061365 is<br><b>ISO</b> . EC 3.1.1.3,<br>sn-1,3 hydrolase.    | Rat, exocrine<br>pancreas/<br>intestinal<br>lumen                                     | High<br>ann<br>fact<br>evid<br>code<br>fetc   |
|  9805004<br>(Carrière<br>1998) | Structural +<br>direct assay<br>(review of<br>primary) | <b>Refutes<br/>PLA1 as<br/>core</b> | Does classical<br>pancreatic<br>lipase have<br>phospholipase<br>A1 activity? | "HPL displays<br>significant<br>activity only on<br>triglycerides,<br>whereas GPLRP2<br>displays high<br>phospholipase<br>and galactolipase<br>activities."<br>Swapping the<br>GPLRP2 mini-lid<br>into HPL did <b>not</b><br>induce<br>phospholipase<br>activity. | Human/<br>guinea-pig<br>pancreatic<br>lipase<br>family;<br>kinetics + 3D<br>structure | High<br>clas<br>dire<br>tran<br>to ra<br>orth |
|  15450178<br>(De Caro<br>2004) | Direct assay                                           | Refutes/<br>Qualifies               | Where does<br>PLA1 reside;<br>how strong?                                    | PLRP2 (paralog)<br>phospholipase<br>activity on PL<br>micelles "very<br>low";<br>galactolipase is its                                                                                                                                                             | Human + rat<br>PLRP2,<br>pancreatic<br>juice                                          | High<br>conf<br>para<br>iden<br>rat           |

| Citation                                                                         | Evidence type           | Supports/<br>Refutes/<br>Qualifies    | Claim tested                                  | Key finding                                                                                                                                                                       | Context                       | Con<br>&<br>limi                            |
|----------------------------------------------------------------------------------|-------------------------|---------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------|
|                                                                                  |                         |                                       |                                               | main function.<br>Antibodies cross-react with native rat PLRP2.                                                                                                                   |                               |                                             |
| <a href="#">P 9805004</a><br>(Carrière 1998)                                     | Structural              | Qualifies                             | Mechanistic basis                             | Full-length ~23-residue lid of classical pancreatic lipase sterically blocks large/polar head-group substrates; GPLRP2 has a 5-residue mini-lid.                                  | Pancreatic lipase gene family | Higl                                        |
| <a href="#">P 37062468</a> / <a href="#">22835523</a> / <a href="#">21554982</a> | Direct assay            | Competing (assigns PLA1 to paralog)   | Which family member is the PLA1/galactolipase | PLRP2 "exhibits remarkable galactolipase and phospholipase A1 activities."                                                                                                        | Guinea-pig/human PLRP2        | Higl<br>suppl<br>para<br>reas               |
| <a href="#">P 32034094</a><br>(Kristensen 2020)                                  | Direct assay/structural | <b>Refutes LPL activity for Pnlip</b> | Is lipoprotein lipase the same activity?      | LPL binds GPIHBP1 and hydrolyzes <b>lipoprotein-bound</b> TG at the capillary endothelium, regulated by ApoC-II/ANGPTL. Distinct enzyme (gene <i>Lpl</i> ), distinct compartment. | Human LPL, intravascular      | Higl<br>esta<br>LPL<br>sepa<br>spec<br>enzy |
| <a href="#">P 32562799</a><br>(ApoC2)                                            | Mutant/model            | Qualifies                             | LPL physiology                                | LPL-mediated plasma TG hydrolysis is                                                                                                                                              | Hamster/rodent                | Med<br>cont<br>supp                         |

| Citation                                                                                                                                                              | Evidence type | Supports/<br>Refutes/<br>Qualifies                                                         | Claim tested                                                        | Key finding                                                                                                                                                                                                                                                                                                                          | Context                                           | Con<br>&<br>limi                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------|
|                                                                                                                                                                       |               |                                                                                            | distinct from<br>digestion                                          | activated by<br>ApoC-II; a plasma/<br>endothelial<br>system unrelated<br>to luminal fat<br>digestion.                                                                                                                                                                                                                                |                                                   |                                      |
| UniProt<br>FUNCTION +<br>CATALYTIC<br>ACTIVITY<br>(P27657)                                                                                                            | Database      | <b>Refutes FA<br/>biosynthesis</b>                                                         | Direction of<br>reaction                                            | "splits the esters<br>of long-chain<br>fatty acids at<br>positions 1 and 3,<br>producing mainly<br>2-<br>monoacylglycerol<br>and free fatty<br>acids"; reaction =<br>hydrolysis<br>(catabolism).                                                                                                                                     | Rat                                               | Higl<br>bios<br>is th<br>opp<br>dire |
|  8656075<br>(Jennens &<br>Lowe 1995, <i>J<br/>Lipid Res</i> ) —<br><b>verified</b> | Direct assay  | <b>Supports<br/>"poor PLA1"<br/>(rat-<br/>specific) +<br/>Refutes<br/>PLA1 as<br/>core</b> | Residual<br>phospholipid<br>hydrolysis;<br>direction of<br>activity | "rPLRP2 could<br>hydrolyze<br>phospholipids, a<br><b>substrate poorly<br/>hydrolyzed by<br/>PL.</b> " rPLRP2<br>(paralog) lacks<br>interfacial<br>activation, is not<br>bile-salt inhibited,<br>weak colipase<br>dependence —<br>opposite of<br>classical PL. Also<br>the <b>IDA source<br/>for GO:0004806<br/>(core TG-lipase).</b> | Rat;<br>recombinant<br>rPLRP2 vs<br>rPL, kinetics | Higl<br>orth<br>prin<br>data<br>own  |

| Citation                                                                       | Evidence type              | Supports/<br>Refutes/<br>Qualifies          | Claim tested                            | Key finding                                                                                                                                                                                                                                                             | Context                               | Con<br>&<br>limi |
|--------------------------------------------------------------------------------|----------------------------|---------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------|
| QuickGO<br>with/from<br>provenance<br>(P27657) —<br><b>computed<br/>in-run</b> | Computational/<br>database | <b>Refutes<br/>(paralog<br/>carry-over)</b> | Origin of the 4<br>flagged IBA<br>terms | All 4 flagged<br>terms = PAINT<br>(GO_REF:0000033)<br>at node<br><b>PTN000906454</b> ,<br>propagated from<br>LPL (P06858),<br>bovine LPL<br>(P11151), hepatic<br>lipase LIPC<br>(P11150),<br>endothelial lipase<br>LIPG (Q9Y5X9).<br>Core TG-lipase<br>term is IDA/EXP. | GO/<br>PANTHER<br>annotation<br>graph | Higl<br>API      |

## GO Curation Implications (leads — require curator verification)

| GO term                                                        | Aspect | Current evidence                                                | Lead action                                                                                                                                                          |
|----------------------------------------------------------------|--------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GO:0004806 triacylglycerol lipase activity                     | MF     | IDA:RGD                                                         | <b>Retain — core MF</b>                                                                                                                                              |
| GO:0016298 lipase activity                                     | MF     | IDA:RGD                                                         | Retain (parent; core)                                                                                                                                                |
| GO:0005576 extracellular region                                | CC     | IDA:RGD                                                         | <b>Retain — core CC</b>                                                                                                                                              |
| GO:0019433 triglyceride catabolic process                      | BP     | IBA                                                             | Retain — correct-direction core process                                                                                                                              |
| GO:0008970 glycerophospholipid phospholipase A1 activity       | MF     | IBA                                                             | <b>Do not treat as core / consider NOT-supported.</b> Refuted by <a href="#">P 9805004</a> ; belongs to paralog PLRP2. At most weak in-vitro sn-1 activity.          |
| GO:0004465 lipoprotein lipase activity                         | MF     | IBA                                                             | <b>Remove / do not propagate.</b> Wrong paralog (gene <i>Lpl</i> ); distinct compartment & cofactors.                                                                |
| GO:0034375 HDL particle remodeling                             | BP     | IBA                                                             | <b>Remove.</b> Function of hepatic/ endothelial lipase, LCAT, CETP, PLTP — not luminal digestive lipase.                                                             |
| GO:0006633 fatty acid biosynthetic process                     | BP     | IBA                                                             | <b>Remove — direction error.</b> Pnlip releases FA by hydrolysis (catabolism), it does not synthesize FA.                                                            |
| GO:0061365 positive regulation of triglyceride lipase activity | BP     | ISO:RGD ← mouse<br>ISO ← human IDA<br><a href="#">P 9631512</a> | <b>Remove / re-map to GO:0004806.</b> Source paper characterizes the enzyme's own colipase-dependent interfacial activation, not regulation of a separate TG lipase. |

Avoid "protein binding" as a recommendation; the informative core MF is **triacylglycerol lipase activity (GO:0004806)**.

**Consolidated GO decision table** (all evidence codes fetched live; saved as artifact `go_decision_table_Pnlip_P27657.csv`):

| GO_ID      | Term                                                | Aspect | Evidence provenance                   | Curator lead                                        |
|------------|-----------------------------------------------------|--------|---------------------------------------|-----------------------------------------------------|
| GO:0004806 | triacylglycerol lipase activity                     | MF     | IDA<br>EXP<br>P 8656075<br>P 10769148 | RETAIN (core)                                       |
| GO:0016298 | lipase activity                                     | MF     | IDA<br>P 15181189<br>8203536          | RETAIN                                              |
| GO:0005576 | extracellular region                                | CC     | IDA<br>P 17010228<br>8967484          | RETAIN (core)                                       |
| GO:0019433 | triglyceride catabolic process                      | BP     | IBA (incl. PLRP2 P54317)              | RETAIN                                              |
| GO:0008970 | phospholipase A1 activity                           | MF     | IBA PTN000906454<br>← LPL/LIPC/LIPG   | REMOVE/non-core (refuted)<br>P 8656075<br>P 9805004 |
| GO:0004465 | lipoprotein lipase activity                         | MF     | IBA PTN000906454<br>← LPL, bovine LPL | REMOVE (gene <i>Lpl</i> )                           |
| GO:0034375 | HDL particle remodeling                             | BP     | IBA PTN000906454<br>← LIPC, LIPG      | REMOVE (HL/EL/LCAT)                                 |
| GO:0006633 | fatty acid biosynthetic process                     | BP     | IBA PTN000906454<br>← LPL, LIPC       | REMOVE (direction error)                            |
| GO:0061365 | positive regulation of triglyceride lipase activity | BP     | ISO ← ISO ← human IDA<br>P 9631512    | REMOVE/RE-MAP to GO:0004806                         |
| GO:0047376 |                                                     | MF     | ISS/IEA/ISO                           |                                                     |

| GO_ID      | Term                                  | Aspect | Evidence provenance | Curator lead             |
|------------|---------------------------------------|--------|---------------------|--------------------------|
|            | all-trans-retinyl-palmitate hydrolase |        |                     | NON-CORE, keep w/ caveat |
| GO:0050253 | retinyl-palmitate esterase activity   | MF     | ISO (MGI)           | NON-CORE                 |
| GO:0030299 | intestinal cholesterol absorption     | BP     | ISO (MGI)           | KEEP w/ caveat           |
| GO:0042632 | cholesterol homeostasis               | BP     | IBA                 | NON-CORE                 |

## Mechanistic Scope

**Direct molecular function (the thing being tested):** interfacial hydrolysis of the sn-1 and sn-3 ester bonds of emulsified dietary triacylglycerols, yielding 2-monoacylglycerol + free fatty acids, in the intestinal lumen, requiring colipase and bile salts, with the amphipathic lid controlling substrate access. This is a **carboxyl-ester hydrolase / catabolic** activity.

**Separated from the seed's secondary claims:** - *Phospholipase A1* — a positional descriptor (sn-1 acyl release) that, for classical pancreatic lipase, is at best a trace in-vitro side reaction; the physiological PLA1/galactolipase is the paralog PLRP2. Not a downstream phenotype — it is a **misassigned molecular activity**. - *Lipoprotein lipase activity / HDL remodeling* — activities of **different genes/enzymes** (LPL, hepatic/endothelial lipase) in the intravascular compartment; not properties of the pancreatic gene product. - *Fatty-acid biosynthesis* — a **pathway/direction confusion**: FA release by hydrolysis  $\neq$  FA biosynthesis. - *Positive regulation of TG lipase* — a **regulatory-role assignment** that conflates the enzyme with a regulator of the enzyme.

## Conflicts and Alternatives

1. **Paralog confusion (primary explanation).** PLA1, LPL, HDL-remodeling annotations are correct for *other* members of PANTHER PTHR11610 (LPL, hepatic lipase, endothelial lipase,

PLRP2). IBA/PAINT propagation from the shared ancestral node (PTN000906454) carried these down to classical pancreatic lipase.

2. **In-vitro-only / positional artifact.** Because pancreatic lipase is sn-1,3 specific, phospholipid assays can register residual sn-1 cleavage; converting this weak, non-physiological turnover into a functional PLA1 annotation over-states it. Equally, one must not falsely convert it to a categorical zero — the accurate statement is "very poor/negligible."
  3. **Direction/term error.** "Fatty acid biosynthetic process" is anabolic; the enzyme is catabolic. This is a term-selection error rather than a paralog issue.
  4. **Orthology carry-over (ISO).** GO:0061365 and the retinyl-ester terms come from ISO to mouse/human RGD orthologs; their applicability to the *direct* rat gene-product function should be checked at source.
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## Knowledge Gaps

| Gap                                                                                | What was checked                                                                             | Why it matters                                                                                                                          | What would resolve it                                                                   |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <p>~~</p> <p><b>P</b> 8656075</p> <p>wording~~</p> <p><b>RESOLVED (Iter 2)</b></p> | <p>Fetches full abstract via NCBI E-utilities</p>                                            | <p>It is the seed's anchor; now verified rat-specific ("poorly hydrolyzed by PL") and is the IDA source for the core TG-lipase term</p> | <p>Done — quantitative kcat/Km on PC vs TG would further quantify the residual rate</p> |
| <p>~~Rat-specific PLA1 measurement~~</p> <p><b>Largely resolved (Iter 2)</b></p>   | <p><b>P</b> 8656075</p> <p>is rat data (rPL vs rPLRP2)</p>                                   | <p>Removes the human/guinea-pig extrapolation concern</p>                                                                               | <p>A modern purified rat PNLIP PC-monolayer assay would give an exact residual rate</p> |
| <p>~~Source of GO:0061365~~</p> <p><b>RESOLVED (Iter 3)</b></p>                    | <p>Traced orthology chain: rat ISO ← mouse ISO ← human PNLIP IDA</p> <p><b>P</b> 9631512</p> | <p>Shows it is a mis-mapping of the enzyme's own interfacial activation, not a genuine regulatory role</p>                              | <p>Done — curator to decide remove vs re-map to GO:0004806</p>                          |
| <p>~~GO:0008970 non-IBA support in rat~~</p> <p><b>RESOLVED (Iter 2)</b></p>       | <p>QuickGO with/from: only IBA from LPL/LIPC/LIPG at PTN000906454</p>                        | <p>Confirms no experimental support; safe to treat as non-core</p>                                                                      | <p>None needed for curation; optional direct assay</p>                                  |

## Discriminating Tests

1. **pH-stat / monomolecular-film kinetics** on purified rat PNLIP vs rat PLRP2 using triolein, egg-PC (PLA1 substrate), and MGDG (galactolipid), ± colipase ± bile salts — expect PNLIP active only on triolein.
2. **Lid-domain swap** (rat PNLIP full lid ↔ PLRP2 mini-lid) to test whether phospholipase activity tracks the lid, replicating

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in the rat background.

3. **Compartment/cofactor test** for the LPL claim: assay PNLIP on lipoprotein-bound TG  $\pm$  ApoC-II  $\pm$  GPIHBP1 — expect no LPL-like activation.
4. **Annotation provenance audit**: pull the PAINT record for PTHR11610 and confirm the ancestral node from which GO:0008970/0004465/0034375/0006633 were propagated (PTN000906454), plus the with/from evidence for GO:0061365.
5. **Positional-product analysis**: if any phospholipid hydrolysis is seen, confirm sn-1 specificity and quantify rate relative to TG (expect <1–5%).

## Curation Leads (require curator verification)

### • Candidate references to cite/verify:

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(verified) — "rPLRP2 could hydrolyze phospholipids, **a substrate poorly hydrolyzed by PL.**" Rat-specific; also the IDA basis for core GO:0004806. (supports weak-not-zero; assigns phospholipid activity to paralog rPLRP2)

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**P** 9805004

— "HPL displays significant activity only on triglycerides, whereas GPLRP2 displays high phospholipase and galactolipase activities" and "The phospholipase activity is, however, not induced in the case of the HPL mutant with GPLRP2 mini-lid." (refutes core PLA1)

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**P** 15450178

— "The phospholipase activity of HPLRP2 on phospholipid micelles was very low" (paralog carries only weak PLA1; confirms rat PLRP2 identity)

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**P** 32034094

— LPL/GPIHBP1 endothelial, lipoprotein-bound-TG mechanism (refutes LPL activity assignment to Pnlip)

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**P** 9631512

(Yang & Lowe 1998) — root IDA for GO:0061365 on human PNLIP: "it was inhibited by bile

salts, required colipase for activity, and demonstrated interfacial activation." Characterizes the enzyme's OWN activity → supports remove/re-map of GO:0061365.

- **QuickGO provenance (computed):** the 4 flagged IBA terms all propagate from node PTN000906454 seeded by LPL (P06858/P11151), hepatic lipase LIPC (P11150) and endothelial lipase LIPG (Q9Y5X9) — direct evidence of vascular-lipase paralog carry-over.
- **Candidate GO actions:** treat **GO:0008970**, **GO:0004465**, **GO:0034375**, **GO:0006633** as **non-core / candidate-remove** for Pnlip (IBA family carry-over; PLA1 & FA-biosynthesis additionally refuted/direction-wrong). Verify **GO:0061365** at source. **Retain** GO:0004806 (core MF), GO:0005576 (core CC), GO:0019433 (correct catabolic BP).
- **Suggested questions for curator:** Is any of the flagged terms supported by non-IBA evidence in rat/human? Should PLA1 be reassigned to *Pnliprp2*? Is GO:0006633 a mis-selected term for a hydrolase?
- **Suggested experiments:** the discriminating tests above (film kinetics; lid swap; LPL-cofactor assay).

## Provenance

Computed in-run: (1) live UniProt P27657 GO/EC fetch + GO-triage table; (2) NCBI E-utilities fetch of

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full abstract (verified rat-specific "poorly hydrolyzed by PL"); (3) QuickGO annotation API for P27657 returning evidence codes, references and with/from seed proteins for every term; (4) UniProt identity resolution of the PTN000906454 with/from accessions (P06858=human LPL, P11151=bovine LPL, P11150=human hepatic lipase LIPC, Q9Y5X9=human endothelial lipase LIPG, P54317=human PNLIPRP2, P16233=human PNLIP); (5) GO:0061365 orthology-chain trace (rat P27657 ISO → mouse Q6P8U6 ISO → human P16233 IDA

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+ E-utilities fetch of

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showing it characterizes the enzyme's own interfacial activation; (6) consolidated GO decision table saved as `go_decision_table_Pnlip_P27657.csv`. Literature verified via PubMed abstracts (PMIDs above). All database facts fetched live; no values fabricated.

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